

Crime Pattern Analysis Website Guidance

Cross-City Spatiotemporal Crime Pattern Prediction with Transfer Learning

English User Guidance for the Interactive Crime Prediction Map

Primary Website Link
Main GitHub repository: https://github.com/Tiffanyxxx3238/spatiotemporal-crime-transfer-learning Main map file: outputs/maps/crime_map_v8.html Direct GitHub file link: https://github.com/Tiffanyxxx3238/spatiotemporal-crime-transfer-learning/blob/main/outputs/maps/crime_map_v8.html Raw HTML download link: https://raw.githubusercontent.com/Tiffanyxxx3238/spatiotemporal-crime-transfer-learning/main/outputs/maps/crime_map_v8.html

Clickable links: [Repository](#) | [crime_map_v8.html on GitHub](#) | [Raw HTML](#)

Document purpose: This guidance explains how to open, operate, explain, and troubleshoot the crime prediction website. It is written for teachers, project evaluators, team members, and first-time users who need a clear walkthrough of the system.

Item	Description
Website type	Standalone interactive HTML map using HTML, CSS, JavaScript, and Leaflet.js
Main file	outputs/maps/crime_map_v8.html
Project goal	Visualize cross-city grid-level crime pattern predictions and support police/public safety interpretation
Main users	Police users, public users, teachers/evaluators, and project presenters
Core outputs	Prediction grids, confidence display, risk score, patrol priority list, route risk query, and safety information

1. Project and Website Overview

This website visualizes the results of a cross-city spatiotemporal crime pattern prediction system. Instead of only presenting model scores in tables, the website converts prediction results into an interactive map. Users can select cities, time slots, seasons, and months, then inspect predicted dominant crime categories at the grid level.

- The project uses multiple cities and standardizes different crime datasets into common categories.
- The prediction task is grid-level, not single-event prediction. Each grid cell represents a small geographic area.
- The map is designed as a decision-support interface, not as a final policing decision-maker.
- The website contains two role-based views: Police View and Public View.

Concept	Plain-English explanation
Cross-city prediction	The system compares and transfers crime patterns across different cities instead of studying only one city.
Spatiotemporal analysis	The model considers both location and time, including time slots, months, and seasons.
Dominant crime category	For each grid, the system predicts the most likely main category: violent, property, or other.
Interactive map	The prediction result is shown with colored grid cells, confidence opacity, and clickable details.

2. Opening the Website

The main map file is stored inside the GitHub repository under the path `outputs/maps/crime_map_v8.html`. Because the map is a large standalone HTML file, the most stable way to demonstrate it is to download the HTML file and open it locally in a modern browser such as Chrome or Edge.

Method	Steps	Recommended for
Open from GitHub	Go to the repository link, open <code>outputs/maps</code> , and select <code>crime_map_v8.html</code> . GitHub shows the file page, but it may not fully run the interactive map.	Finding the file path or showing the project repository
Download raw HTML	Open the Raw HTML link, save the file as <code>crime_map_v8.html</code> , then open it with Chrome or Edge.	Stable classroom demo
Open local HTML	Double-click the downloaded HTML file or right-click and choose Open with Chrome / Edge.	Offline or quick demo
Host locally	Put the HTML in a folder, run <code>python -m http.server 8000</code> , then open <code>http://localhost:8000/crime_map_v8.html</code> .	More stable browser behavior and device testing

Important Demo Tip

If GitHub does not directly render the interactive map, this is normal. GitHub is mainly for storing the file. For the live interactive demo, download the raw HTML file and open it in a browser, or serve it through a local HTTP server.

3. First Screen: Intro Animation and Role Selection

When users enter the website, they first see a short intro animation for the crime pattern analysis system. After the loading/intro screen, the website asks the user to choose a view. This role selection controls how much information is shown and how the results are explained.

View	Target user	What it emphasizes
Police View	Police officers, patrol planners, city safety managers	High-risk grids, risk scores, patrol ranking, alert threshold, and patrol check-in
Public View	General public or non-technical users	Safety summaries, simpler explanations, lower-risk areas, and daily route guidance

Recommended explanation during presentation:

“The same prediction map can adjust its information density for different users. Police View focuses on patrol planning and decision support, while Public View focuses on simple safety information and avoids overly technical or enforcement-oriented details.”

4. Main Interface Layout

After choosing a view, the main interface is divided into a sidebar and an interactive map area. The sidebar contains the city buttons, time controls, filters, analysis tools, and role-specific panels. The map area displays colored prediction grids and interactive geographic information.

Interface area	Function
Sidebar header	Shows the website title and current project identity.
City tabs	Switch between the available cities in the map dataset.
Time slider	Select one of four time slots: Night, Morning, Afternoon, or Evening.
Auto Play button	Automatically cycles through time slots to show temporal changes.
Season / Month filters	Filter the prediction map by season or a specific month.
Map panel	Displays prediction grids, alert markers, and clickable grid details.

Role-specific panel	Shows patrol guidance for Police View or safety guidance for Public View.
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5. City Selection

The map supports multiple cities. Users can select a city from the city tabs in the sidebar. After switching cities, the map recenters and updates the grids, statistics, and role-specific information.

Available cities in the v8 map

NYC, Chicago, LA, London, Philadelphia, DC, West Yorkshire, Detroit, Cambridge, Salt Lake City, Birmingham, Peoria, Kansas City, Karachi, Dallas, San Francisco, Seattle

- Use large cities such as NYC, Chicago, LA, and London to show dense prediction grids and richer spatial patterns.
- Use smaller cities such as Cambridge, Peoria, Birmingham, or Salt Lake City to explain how the system can also support data-poor cities.
- When comparing cities, remind users that risk scores are normalized within each city and should not be treated as absolute cross-city rankings.

6. Time Slot, Season, and Month Controls

Control	How to use it	What it means
Time Slot	Move the time slider or click a time option.	Shows prediction patterns for Night, Morning, Afternoon, or Evening.
Auto Play	Click Auto Play to cycle through the four time slots.	Demonstrates how crime patterns change throughout the day.
Season Filter	Choose All Year, Spring, Summer, Fall, or Winter.	Displays results grouped by seasonal context.
Month Filter	Choose All or a month from January to December.	Displays month-specific prediction patterns.

Presentation sentence: “The Method B design adds month information, so the map can support seasonal and monthly exploration rather than only showing one static yearly result.”

7. Reading the Map Correctly

The colored grid cells are the core of the website. Each grid cell represents a small geographic area, and the color shows the model-predicted dominant crime category. The opacity shows prediction confidence: darker cells mean higher confidence and lighter cells mean lower confidence.

Visual element	Meaning	How to explain it
Red grid	Predicted dominant category is violent crime.	This does not mean a violent crime will definitely occur; it means violent crime is the predicted main category for that grid and time condition.
Blue grid	Predicted dominant category is property crime.	This usually represents theft, burglary, fraud, vehicle theft, or similar property-related patterns.
Green grid	Predicted dominant category is other crime.	This can include lower-sample or mixed categories such as public order, drug, disorder, or other non-violent/non-property patterns.
Darker opacity	Higher calibrated confidence.	The model is more confident in the predicted category.
Lighter opacity	Lower confidence or uncertainty.	Users should interpret the prediction more

		carefully.
Alert marker	Grid exceeds the selected risk threshold.	Used to highlight areas that may require more attention.

Responsible Interpretation

The map predicts dominant crime patterns, not guaranteed future events. The result should support human judgment, patrol planning, and safety analysis, but it should not be used as the only basis for policing decisions.

8. Grid Detail Panel

Users can click a grid cell on the map to inspect more detailed information. This helps connect the visual map to the model output and makes the prediction easier to explain during a presentation.

Detail item	Meaning
Coordinates	The latitude and longitude of the selected grid cell.
True Category	The observed dominant category in the evaluation data, when available.
Prediction	The model-predicted dominant crime category.
Confidence	The calibrated probability or confidence level of the prediction.
Event Count	The number of crime events represented in the grid-time-month sample.
Risk Score	A city-normalized risk score used for within-city comparison.
Category Probabilities	The probability distribution across violent, property, and other categories.

9. Police View Guidance

Police View is designed for patrol planning and operational decision support. It intentionally shows more detailed information than Public View, including patrol priority, threshold presets, high-risk grids, and check-in progress.

Police View feature	How to use it	Why it matters
Patrol Priority List	Open the Patrol section and review the automatically ranked grid list.	Helps officers identify which areas should be checked first.
Risk Threshold	Adjust the threshold slider or use presets such as routine/enhanced/urgent.	Controls how many grids are considered high risk.
Alert Markers	Turn alert markers on/off in the layer controls.	Highlights grids that exceed the selected risk level.
Check In button	Click Check In after a grid has been visited.	Demonstrates patrol progress and covered areas.
Progress bar	Review how many listed grids have been checked.	Shows patrol completion during the current session.
Clear check-ins	Click Clear to reset the current session.	Useful before starting a new demo or patrol scenario.

Suggested police-view explanation:

“Police View helps convert model predictions into patrol guidance. The system ranks higher-risk grids, shows the predicted crime type, displays confidence and risk score, and allows officers to check in after visiting a grid. At this stage, check-in is a prototype state stored during the current browser session.”

10. Public View Guidance

Public View is simplified so that non-technical users can understand the map without being overwhelmed by model details. It is meant to provide safety awareness, not to create fear or label any area permanently as dangerous.

Public View feature	User action	User benefit
Area summary	Read the summary card for the selected city and time.	Quickly understands the current risk pattern.
Lower-risk areas	Review the lower-risk grid list or click to move the map.	Helps users find relatively safer areas within the selected city.
Safety note	Read the simple guidance text.	Provides daily-use interpretation without technical terms.
Map search / route use	Use route or map navigation tools if needed.	Supports safer travel planning.

Suggested public-view explanation:

“Public View keeps the information simpler. It focuses on safety summaries, lower-risk areas, and daily risk reminders, while avoiding too much enforcement-oriented detail.”

11. Route Risk Query

The route risk query allows users to enter a starting point and a destination. The system checks the prediction grids along the route and summarizes the risk along that path.

1. Open the Analysis section in the sidebar.
2. Find the Route Risk Query area.
3. Enter the start latitude and longitude.
4. Enter the end latitude and longitude.
5. Click Calculate Route Risk.
6. Review route grids, average risk, maximum risk, and category distribution along the route.

Result	Interpretation
Route Grids	How many grid cells the route passes through.
Average Risk	The mean risk score along the route.
Maximum Risk	The highest-risk segment along the route.
Violent Share	The proportion of route grids predicted as violent category, if shown.

12. Language, Theme, and Layer Controls

Control	Purpose
Chinese / English switch	Changes the interface language for bilingual presentation.
Light / Dark mode	Switches the visual theme for different classroom or screen conditions.
Prediction layer	Shows or hides the main prediction grid cells.
Violence heatmap	Shows or hides the violence-focused heat layer when available.
Alert marks	Shows or hides warning markers above the selected threshold.
Choose View	Returns to role selection or switches between Police View and Public View.

13. Suggested Presentation Walkthrough

Use the following sequence for a clear live demonstration. This order starts from the project purpose, then shows how the interface works, and finally connects the website to the model and real use cases.

7. Open the GitHub repository and point out the main file path: `outputs/maps/crime_map_v8.html`.
8. Open the downloaded/local HTML map in Chrome or Edge.
9. Explain the intro animation and say the website visualizes cross-city crime predictions.
10. Choose Police View first to show the more complete operational interface.
11. Select a major city such as NYC or Chicago and explain that the city is divided into prediction grids.
12. Use the time slider and Auto Play to show how patterns change across Night, Morning, Afternoon, and Evening.
13. Switch season or month filters to explain the monthly/seasonal design.
14. Explain colors: red means violent, blue means property, green means other; opacity means confidence.
15. Click one grid and explain coordinates, prediction, confidence, event count, and risk score.
16. Open Patrol and show the ranked patrol list, threshold, Check In button, and progress bar.
17. Switch to Public View and show how the information becomes simpler and more safety-oriented.
18. Use Route Risk Query to demonstrate how the system can support route awareness.
19. Close by emphasizing that the map is a decision-support tool, not a final decision-maker.

14. Short English Speaking Script

Opening:

"This is our cross-city crime prediction map website. It visualizes the model output through an interactive Leaflet map, so users can understand crime patterns by city, time slot, season, and month."

Map explanation:

"Each grid cell shows the predicted dominant crime category. Red means violent crime, blue means property crime, and green means other crime. The opacity represents calibrated confidence, so darker cells indicate higher confidence."

Police View:

"Police View is designed for patrol planning. It ranks high-risk grids, provides risk scores and predicted crime types, and includes a prototype check-in function to mark visited patrol locations."

Public View:

"Public View is designed for general users. It keeps the interface simpler and focuses on safety summaries, lower-risk areas, and everyday risk guidance."

Closing:

"The system combines machine learning, cross-city transfer learning, and geographic visualization. It does not replace human judgment, but it can support safer patrol planning and public safety awareness."

15. Technical Background for Evaluators

Layer	Technology / design	Function
Data layer	Multi-city crime datasets	Collects and standardizes crime records from different cities.
Feature layer	Grid-time-month samples and 26 feature groups	Transforms raw records into model-ready tabular features.
Model layer	CatBoost, LightGBM, RandomForest, XGBoost, calibration, transfer learning	Predicts dominant crime categories and confidence.
Visualization layer	HTML, CSS, JavaScript, Leaflet.js	Displays predictions as an interactive map.
User layer	Police View and Public View	Adapts information density for different users.

- The map is a standalone demonstration version. It can run without a backend server because the prediction data is embedded in the HTML file.
- The current check-in feature is a prototype. It is stored in the current browser session and does not permanently write to a database.
- For formal deployment, future work could add login, backend database storage, patrol history, officer ID, timestamps, and exportable reports.

16. Troubleshooting

Problem	Possible reason	Solution
GitHub opens the code instead of an interactive map	GitHub is showing the file page, not running the HTML app.	Download the Raw HTML file and open it locally in Chrome or Edge.
The page loads slowly	The HTML is large because it contains embedded prediction data.	Wait for loading to finish, use Chrome/Edge, or run it through a local server.
Map tiles do not appear	Internet connection or map tile loading issue.	Check internet connection and refresh the page.
Check-in progress disappears after refresh	Current version stores check-ins in browser memory only.	This is expected for the prototype; future version can use a database.
The map looks too dark or too light	Theme setting or room projector brightness.	Use the Light/Dark toggle.
Too many or too few alert markers	Risk threshold is too low or too high.	Adjust the threshold slider or use presets.

17. Limitations and Responsible Use

The website should be presented carefully because crime prediction systems can be misunderstood. The correct positioning is decision support. It should help users observe patterns, compare areas, and plan patrols, but it should not be treated as an absolute prediction of future criminal events.

Limitation	Explanation
Prediction is pattern-level	The model predicts dominant crime category by grid, time slot, and month. It does not predict an exact future crime event.
Data quality varies by city	Different cities use different formats, crime definitions, and reporting systems.
Risk score is city-normalized	Risk scores are useful within the same city but should not be used as direct absolute comparisons across cities.
Historical bias may exist	Crime records reflect reporting and enforcement patterns, so results must be interpreted responsibly.
Prototype features are not permanent	Check-in and some interface states may reset when the page is refreshed.

18. Quick Checklist for Live Demo

- Open the local HTML file before class and make sure the map loads correctly.
- Prepare one city example, such as NYC or Chicago, to explain city switching and time slots.
- Prepare one Police View example to show patrol ranking and check-in.
- Prepare one Public View example to show safety summary and lower-risk areas.
- Prepare one route query example if you want to demonstrate route risk analysis.
- Remind the audience that this is a decision-support tool, not a guarantee of crime occurrence.

19. Reference Links

- GitHub repository: <https://github.com/Tiffanyxxx3238/spatiotemporal-crime-transfer-learning>
- Main map file: outputs/maps/crime_map_v8.html: https://github.com/Tiffanyxxx3238/spatiotemporal-crime-transfer-learning/blob/main/outputs/maps/crime_map_v8.html
- Raw HTML download link: https://raw.githubusercontent.com/Tiffanyxxx3238/spatiotemporal-crime-transfer-learning/main/outputs/maps/crime_map_v8.html

Recommended file path to mention in reports: outputs/maps/crime_map_v8.html